


# Stress Testing

User Guide — CDS-in-Stress, Flood Classifiers & Portfolio Scenarios

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# 1 Introduction

Stress testing answers the question: *what would happen to our PRS positions if a specific storm hit the catchment?*

The platform uses its catalogue of 20,000 synthetic storm sequences to run hourly scenarios. For each hour of a 168-hour storm, the system computes the gauge water level, the probability of flooding, and the resulting P&L impact on every affected trade. This produces a time-resolved view of how the portfolio behaves under stress.

## 1.1 CDS-in-Stress Concept

In normal mark-to-market, PRS positions are valued using CDS-style analytics (hazard rate → fair spread → MTM). Under stress, the position transitions from a credit instrument to a **cash-settled physical payout**:

$$\text{cash price} = N \times P(\text{flood}) \times d$$

where $N$ is the notional, $P(\text{flood})$ is the real-time flood probability, and $d$ is the direction ($+1$ payer, $-1$ receiver). The stress P&L is the difference between this cash price and the normal-market MTM.

# 2 Gauge-Level Stress

## 2.1 Workflow

1. **Select a gauge** from the dropdown. Only gauges with open trades are shown.
2. **Select a storm** from the catalogue. The list shows storms that breached alert level at this gauge, sorted by severity (catastrophic first).
3. **Run the scenario.** The server computes the hourly P&L trajectory over 168 hours.

## 2.2 What the Charts Show

The stress tab displays three sub-charts:

**Flood Probability** A line chart of $P(\text{flood})$ vs. hour. Shows how the probability rises as water level approaches the trigger threshold, peaks, and then recedes.

**Stress P&L** A line chart of portfolio stress P&L vs. hour. Shows the accumulated cash-price impact of the storm.

**Probability Surface** A 3D surface of $P(\text{flood})$ vs. water level vs. hour. Useful for understanding how sensitivity changes at different water levels.

# 3 The Hourly Scenario

For each hour $h$ in the 168-hour window:

1. **Water level** $w_h$ is read from the storm's gauge response hydrograph (pre-computed from the storm catalogue).
2. **Flood probability** $p_h = f(w_h, h)$ is computed using either a trained GBM classifier or the analytical flood polynomial.
3. **Cash price** per trade: $C_h = N \times p_h \times d$.
4. **Stress P&L** per trade: $\Pi_h = C_h - \text{MTM}$.
5. **Portfolio stress P&L**: $\sum \Pi_h$ across all affected trades.

### 3.1 Knock-Out Behaviour

When the water level exceeds a trade's trigger threshold:

- The trade "knocks out" — the protection buyer receives full notional.
- $P(\text{flood}) = 1.0$ from the knock-out hour onward.
- Stress P&L jumps to $N - \text{MTM}$ (full notional minus the original mark).
- The trade is no longer re-priced in subsequent hours.

In the chart, the P&L line flattens after knock-out because no further P&L change can occur.

### 3.2 Hydrograph Source

The hourly water levels come from the gauge response stored in `gaugets/{gauge_id}.json`. The storm's `peak_level_m` and duration are used to scale the hydrograph template. If the gauge time series is not available, the system falls back to a flat hydrograph (constant water level for 168 hours).

## 4 Flood Probability Models

Two models can compute $P(\text{flood})$. The classifier is preferred when available because it captures non-linear effects.

### 4.1 GBM Classifier (Preferred)

A **Gradient Boosting Machine** trained per gauge using historical storm simulations. The model takes four features:

| Feature | Description |
|---|---|
| `log_h_s` | Log of water level relative to severe threshold: $\ln(w/w_{\text{severe}} + \varepsilon)$ |
| `log_t_end` | Log of time remaining in the 168-hour window: $\ln(168 - h + \varepsilon)$ |
| `delta_log_h` | First derivative of log water level (rate of rise/-fall) |
| `delta2_log_h` | Second derivative (acceleration of rise/fall) |

The classifier outputs a probability in $[0, 1]$. It is more accurate than the polynomial model because it learns the non-linear relationship between water level dynamics and flood outcomes from the Monte Carlo data.

### 4.2 Analytical Flood Polynomial (Fallback)

When no classifier is trained for a gauge, the system uses a polynomial approximation:

$$P(\text{flood}) = \sigma\left(a_0 + a_1 \frac{w}{w_s} + a_2 \left(\frac{w}{w_s}\right)^2 + a_3 \frac{h}{168}\right)$$

where $\sigma$ is the logistic sigmoid and the coefficients are calibrated to the Thames. This model is smooth but cannot capture the non-linear dynamics learned by the classifier.

### 4.3 When to Use Which

| | **GBM Classifier** | **Flood Polynomial** |
|---|---|---|
| Accuracy | Higher (trained on data) | Lower (generic fit) |
| Availability | Requires training | Always available |
| Speed | Fast (tree lookup) | Instant (formula) |
| Per-gauge | Yes (gauge-specific) | No (one model for all) |

## 5 Portfolio-Level Stress

The Portfolio Stress tab runs a single storm across **all gauges** simultaneously, computing the aggregate impact.

### 5.1 Workflow

1. The tab lists all storms in the catalogue, sorted by the number of gauges that breached severe (most impactful first).
2. Select a storm.
3. The server runs the scenario for every gauge that has open trades and returns per-gauge results.

### 5.2 Output

For each gauge, the response includes:

- Peak water level and threshold classification (alert, warning, severe, or clean).
- $P(\text{flood})$ at peak.
- Stress P&L contribution.
- Number of affected trades.
- Full 168-hour hydrograph.

The portfolio summary shows:

- Total stress P&L across all gauges.
- Number of gauges at each severity level.
- Worst-hit gauge and its contribution.

### 5.3 Drill-Down

Clicking a gauge in the portfolio stress results switches to the Stress tab (single gauge) and pre-selects that gauge and storm, allowing detailed hourly analysis.

## 6 Probability Surface

The probability surface is a 3D visualisation of $P(\text{flood})$ as a function of water level and hour. It is generated by evaluating the classifier (or polynomial) across a grid of water levels and time points.

### 6.1 How to Read It

- The x-axis is time (0–168 hours).
- The y-axis is water level (metres).
- The z-axis (colour) is $P(\text{flood})$.
- The actual storm trajectory is a path across this surface.

The surface reveals the classifier's learned behaviour:

- At low water levels, $P(\text{flood}) \approx 0$ regardless of hour.
- At high water levels, $P(\text{flood}) \approx 1$ regardless of hour.
- The transition zone (e.g. 0.2–0.8 probability) is where the storm's trajectory determines the outcome.
- Late in the window (hour 140+), the probability decreases because there is insufficient time for a full flood event to develop.

## 7 Classifier Training

### 7.1 Training a Classifier

To train a GBM classifier for a gauge:

1. Navigate to the Classifiers tab.
2. Select the gauge.

3. Click **Train**. The system builds a feature matrix from all storm scenarios at this gauge and trains a GBM model.
4. Training typically takes 10–60 seconds per gauge.
5. The trained model is saved as a `.joblib` file in `classifiers/{gauge_id}.joblib`.

### 7.2 Batch Training

The **Train All** button trains classifiers for all gauges with sufficient historical data. This is an asynchronous job; progress is shown in the UI.

### 7.3 AUC Metric

After training, the classifier's quality is measured by the **Area Under the ROC Curve** (AUC):

$> 0.95$ Excellent — very reliable flood probability estimates.

0.85–0.95 Good — reliable for stress testing.

$< 0.85$ Fair — consider retraining with more data or using the polynomial fallback.

### 7.4 When to Retrain

Classifiers should be retrained when:

- The storm catalogue is regenerated (new storm control parameters).
- Gauge hazard data is updated.
- The AUC drops below 0.85 (indicates the model is stale).

## 8 Model Selection: Challenger Comparison

### 8.1 Open Question

> *Should an alternative classifier, such as a Random Forest, be considered in place of the GBM? What evidence shows that the GBM is the most accurate and most stable choice for $P(\textit{flood})$?*

The GBM was selected on the basis of in-sample AUC. That is not sufficient evidence for model selection: no challenger has been fitted on the same data, and the AUC was measured on a split that overstated out-of-sample performance (Section 8.4). This section sets out the comparison that would settle the question. The protocol is written so that the GBM *can fail*: if a challenger wins under the pre-agreed decision rule, the challenger is adopted.

### 8.2 Why Random Forest Is the Natural Challenger

Both models are tree ensembles on the same four features, so they differ only in how the ensemble is built:

| | **GBM (champion)** | **Random Forest (challenger)** |
|---|---|---|
| Ensemble | Sequential, each tree fits the residual of the last | Independent deep trees on bootstrap samples, averaged |
| Error reduced | Mainly bias | Mainly variance |
| Tuning | Sensitive (learning rate, depth, trees) | Robust to defaults |
| Probabilities | Optimises log loss directly; usually well calibrated | Averaged votes; pulled away from 0 and 1, often needs re-calibration |
| Seed sensitivity | Low–moderate (`subsample=0.8`) | Low with many trees |

Because the classifier output is used directly as a price ($C_h = N \times p_h \times d$), calibration matters more than ranking. This is the main reason to expect the GBM to win, but it has to be shown, not assumed.

### 8.3 Candidate Set

1. **GBM** — production configuration (`n_estimators=300`, `max_depth=3`, `learning_rate=0.05`, `subsample=0.8`, `min_samples_leaf=50`).
2. **GBM, tuned** — the same model given the same tuning budget as the challengers. If it beats the production GBM, the production settings, not the model family, need changing.
3. **Random Forest** — `n_estimators=500`, `max_features=2`, `min_samples_leaf` tuned.
4. **Random Forest, calibrated** — the same, wrapped in isotonic or Platt calibration fitted on an inner fold. This stops RF losing only because of calibration that a one-line fix would remove.
5. **Logistic regression** on the four features — a transparent baseline.
6. **Flood polynomial** — the production fallback, as the floor every model must beat.

Every tree model gets the same tuning budget (for example 30 random-search trials) inside the training folds, so the GBM does not win only because it has been tuned more.

### 8.4 Prerequisite: Remove Storm Leakage

Each storm contributes 168 hourly rows, and the label (*future maximum level* $\geq w_{\text{severe}}$) is shared by adjacent hours. Up to version 1.1 the trainer used a row-level stratified `train_test_split`, so hours from the same storm appeared in both training and test sets, and near-miss rows were scored too. The reported AUC therefore measured interpolation within storms already seen, not prediction

on new storms. This leakage favours deep, low-bias learners and so would flatter the Random Forest more than the GBM.

The production trainer now holds out 20% of storms, stratified on whether the storm floods, keeps near-miss rows in training and records `split` in its result. If no such split is possible (for example, only one storm floods) it falls back to the row-level split and records `split = row`. Classifiers and metrics trained before this change keep the old split until they are retrained, when the fitted models also change. On one test-portfolio gauge the new split moved Brier from 0.0213 to 0.0258 and AUC from 0.988 to 0.990. AUC can hold up because the easy gains from leakage are offset by no longer scoring near-miss rows, so Brier and log loss are the better guide to the change.

For the comparison itself:

- Carry a `storm_id` with every training vector.
- Split by storm using `GroupKFold` (5 folds), so every storm is entirely in train or entirely in test.
- Add near-miss synthetic vectors to the *training* folds only; score on real storm vectors only.

### 8.5 Accuracy Metrics

Scored on held-out storms, per gauge and pooled:

**Primary** Brier score and log loss — proper scoring rules that reward correct probabilities, not just correct ranking.

**Calibration** Reliability diagram, expected calibration error, calibration slope and intercept.

**Discrimination** AUC (secondary, kept for continuity with the existing thresholds).

**Business** Stress P&L error on held-out storms: the hourly $\sum_h N p_h d$ path compared against the realised knock-out outcome of that storm.

### 8.6 Stability Tests

"Most stable" is tested in five ways. Each produces a number per model and per gauge.

1. **Seed stability.** Retrain each model with 20 random seeds on the same data. Measure the standard deviation of $p$ on a fixed grid (the probability surface of Section 4) and of peak stress P&L for a reference storm set.
2. **Data stability.** Retrain on 50 bootstrap resamples *of storms* (not of rows). Report the spread of $p$ on the fixed grid and the spread of the held-out Brier score.
3. **Catalogue stability.** Train on one generation of the storm catalogue and score on a regenerated catalogue (a new seed or new storm-control parameters). This is the retraining trigger in practice, so it is the most decision-relevant stability test. The benchmark tool does not regenerate the catalogue itself: run it once on each catalogue generation and compare the two result files.
4. **Shape stability.** $P(\text{flood})$ should not fall as the water level rises. Count monotonicity violations in `log_h_s` across the grid, and measure how rough the finite-difference sensitivity $\partial p / \partial w$ is. Step-like surfaces give noisy stress Greeks.

5. **Cross-gauge consistency.** The distribution of per-gauge metrics: a model that is excellent on most gauges but fails badly on a few is less stable than one that is uniformly good.

### 8.7 Statistical Comparison

- Per gauge: paired difference in Brier score (GBM minus challenger) on the same held-out storms, with a 95% confidence interval from a storm-level bootstrap.
- Across gauges: Wilcoxon signed-rank test on the per-gauge Brier differences.
- Stability metrics: report ratios (challenger dispersion / GBM dispersion) with bootstrap intervals.

### 8.8 Decision Rule

Agreed before the results are seen. The GBM is retained if all of the following hold:

1. Its pooled Brier score is no worse than the best challenger's by more than a set tolerance (for example 2%), and the Wilcoxon test does not show the challenger significantly better.
2. Its seed, bootstrap and catalogue dispersion are no greater than the challenger's (within 0.002 in probability s.d.).
3. It has no more monotonicity violations than the challenger (within 0.5 percentage points of the violation rate).

If a challenger is better on accuracy but worse on stability (or the reverse), the trade-off is recorded and the choice is escalated for model-risk sign-off rather than decided by the model owner. If the best challenger wins on all three, it replaces the GBM. With fewer than six gauges the Wilcoxon test has no power, so "significantly better" then means every gauge's storm-bootstrap interval favours the challenger.

### 8.9 Running the Benchmark

The comparison is implemented in the benchmark tool:

**Code** `tools/flood_classifier_challenger/`

**Settings** `config/models/_flood_challenger.py`

It builds each gauge's training set with the production vector builder, so it scores exactly the features and labels production trains on:

```
python -m tools.flood_classifier_challenger \
    --gauges all
python -m tools.flood_classifier_challenger \
    --gauges G1,G2 --seeds 20 --bootstraps 50
python -m tools.flood_classifier_challenger \
    --synthetic 3     # no storm data needed
```

Defaults are 1,000 storms per gauge, 5 storm folds, 6 tuning trials, 10 seeds and 20 storm bootstraps; a sign-off run uses 20 seeds and 50 bootstraps as above. Fits run in parallel on every core. The run writes `challenger/challenger_results.json` (every metric, the tuned parameters per fold and the probability surfaces) and the table below.

### 8.10 Results

**Run:** 2026-10-30, source `catalogue thames from PhysicalRisk-testdata`, 6 gauges scored (0 skipped), 500 storms per gauge, 5 storm folds, 6 tuning trials, 10 seeds, 20 storm bootstraps.

| Model | Brier | Log loss | AUC | ECE | Seed s.d. | Boot. s.d. | Mono. viol. |
|---|---|---|---|---|---|---|---|
| GBM (production) | 0.0204 | 0.0688 | 0.989 | 0.0078 | 0.0090 | 0.0353 | 8.3% |
| GBM (tuned) | 0.0191 | 0.0644 | 0.990 | 0.0051 | 0.0000 | 0.0495 | 8.9% |
| Random Forest | 0.0185 | 0.0631 | 0.990 | 0.0054 | 0.0052 | 0.0354 | 11.2% |
| Random Forest (calib.) | 0.0199 | 0.0752 | 0.988 | 0.0061 | 0.0420 | 0.0610 | 5.3% |
| Logistic regression | 0.0378 | 0.1438 | 0.921 | 0.0244 | 0.0000 | 0.0175 | 0.0% |
| Flood polynomial | 0.0203 | 0.0822 | 0.979 | 0.0074 | – | – | 11.5% |

**Decision:** best challenger by pooled Brier is Random Forest (GBM Brier 0.0204 vs 0.0185, gap 10.03%; Wilcoxon $p = 0.016$). Accuracy test failed, stability test failed, shape test passed: the result is **mixed** and is escalated for model-risk sign-off.

| Gauge | Storms | Flood rate | GBM Brier | Best chall. | Δ (95% CI) |
|---|---|---|---|---|---|
| GAUGE-ef7f75ec | 500 | 12.6% | 0.0297 | 0.0270 | +0.0027 [+0.0015, +0.0039] |
| GAUGE-066fc60e | 500 | 7.3% | 0.0188 | 0.0166 | +0.0022 [+0.0010, +0.0034] |
| GAUGE-bc801f07 | 500 | 3.6% | 0.0111 | 0.0102 | +0.0009 [+0.0001, +0.0018] |
| GAUGE-3324d948 | 500 | 7.3% | 0.0210 | 0.0189 | +0.0022 [+0.0010, +0.0032] |
| GAUGE-acc1b9cc | 500 | 6.5% | 0.0208 | 0.0197 | +0.0011 [+0.0001, +0.0022] |
| GAUGE-a03e456e | 500 | 7.1% | 0.0208 | 0.0187 | +0.0021 [+0.0008, +0.0033] |

Δ is GBM minus challenger per-storm Brier; positive favours the challenger.

**Reading these results.** This run is illustrative, not sign-off evidence: it uses the first six gauges of the 18 in the test portfolio, at 500 storms each and the default seed and bootstrap counts. The tuned GBM's zero seed s.d. does not show better stability: tuning chose `subsample` = 1 for the fold used in the stability test on all six gauges, so the fit has no randomness for the seed to act on. Its bootstrap s.d., which measures sensitivity to the storms sampled, is the higher of the GBM variants and above the Random Forest's. Tuned parameters also differ between folds, so its accuracy is not that of one fixed setting.

## 9 Parameter Reference

| Parameter | Default | Increase | Decrease |
|---|---|---|---|
| STRESS_STORMS_MIN_COUNT | 50 | Excludes gauges with few storms | Includes more gauges |
| STORM_DEFAULT_DURATION | 168 h | Longer fallback hydrograph | Shorter fallback |
| DEFAULT_PEAK_POSITION | 0.5 | Late peak (delayed climax) | Early peak (front-loaded) |
| LOG_END | 168.0 | Classifier feature scaling | Narrower time window |
| LOG_EPS | $10^{-8}$ | Less numerical sensitivity | More sensitivity at zero |
| BUMP_1BP | 0.0001 | Larger sensitivity step | Finer stress resolution |

## Document History

| Version | Date | Author | Description |
|---|---|---|---|
| 1.0 | 10-Apr-2026 | MKM Research Labs | Initial release |
| 1.1 | 29-Oct-2026 | MKM Research Labs | Added GBM challenger comparison |
| 1.2 | 30-Oct-2026 | MKM Research Labs | Trainer holds out whole storms |